A Comparative Study of Yolo Architectures for Volume-integrated Stock Pattern Recognition: Empirical Evidence from The Vietnamese Market

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Abstract. The financial markets are always chaotic and unstable. The accurate identification of reversal candlestick patterns is the key of technical analysis. This research build a framework for the automated detection of four famous patterns: Head and Shoulders, Inverse Head and Shoulders, Double Top, and Double Bottom. With the data in the period of 2015 -2026 collected from VN30, we generate a dataset that integrates both price action and trading volume, a methodology proved by John Murphy's technical analysis principles. We conducted a comparative analysis by training three architectures from scratch: YOLOv8n, YOLO11n, and YOLO26n. Empirical results show that YOLOv8n offers a good balance for financial object detection, achieving a peak mAP@0.5 of 85.6%, with a notably high precision of 88.3% for the challenging Double Top class. The YOLO11n presents competitive results while YOLO26n prioritizes prediction speed at the expense of detection accuracy. The study's primary contribution is based on the successful synergy between computer vision and domain financial knowledge.

Keywords: Deep Learning, YOLO, Candlestick Pattern Recognition, Quantitative Finance, Technical Analysis, Object Detection, Trading Volume Confirmation

1 Introduction

Nowadays, in the stock market field, predicting stock price movements precisely is still a major challenge, mostly because market data are highly volatile, non-linear, and influenced by random variables. According to John Murphy's study in stock market technical analysis [1], price changes are not only numerical changes but also reflect the collective psychology and behavior of market participants. His study said that price chart patterns can signal potential trend reversals or continuations. Moreover, Murphy said that a pattern without significant volume support often lacks predictive power. However, it is often considered subjective when identifying the patterns because different analysts have different points of view about the charts. Because this bias often creates noise, we must seek new methods that are more objective and stable.

Recognizing this need, there has been increasing research on automated chart-pattern recognition systems based on Computer Vision (CV) to reduce human bias and improve reliability [2, 3].

The diversity of the Vietnamese economy creates a highly volatile stock market, resulting in complex price charts. It's also difficult to manually track hundreds of stocks. Therefore, recent research [4, 5] suggests that as the market develops, applying advanced and modern Deep Learning (DL) models is essential to overcome the limitations of traditional analytical methods. This would allow modern stock analysts to better understand the complex characteristics and patterns of the Vietnamese stock market.

Convolutional Neural Net-works (CNNs) is one of the most important developments in addressing the two issues mentioned above. While traditional analytical algorithms process stock price data as one-dimensional vectors, CNNs models can read stock price charts and process them as two-dimensional vectors. From there, the model can read the characteristics of the price chart in general and stock patterns in particular. The model will learn important information that makes up those patterns, such as edges, peaks, slopes, and curves [6]. And from there it can identify the patterns such as a "Head and Shoulders" or a "Double Bottom”.

One limitation of current Computer Vision studies in the stock market is their heavy reliance on price charts. Murphy's core principle of technical analysis said that stock patterns must be supported by trading volume to be meaningful and predictive [1]. To address this, we offer an enhanced method that integrates price charts and trading volume. Instead of modeling trading volume as a separate numerical input, we combine volume and price charts into one image and then are processed by CNNs. This design enables the YOLO model to learn the geometric structure of price patterns and trading volumes at the same time, leading to more reliable pattern recognition.

This paper has five sections. Section 2 provides a theoretical foundation of chart patterns and recent YOLO research in finance pattern recognition. Section 3 describes the detailed pipeline of our proposed Stock Pattern Prediction framework from data acquisition to labeling. Section 4 presents the evaluation of YOLOv8n, YOLO11n, and YOLO26n and forecasts in the Vietnam stock market through early 2026. The final concludes by summarizing our contributions, identifying limitations during validation, and suggested future improvements to advance the field.

2 Literature Review

2.1 Foundations of Technical Chart Patterns

Technical analysis is not only based on single candlesticks but also focuses on complex price structures known as chart patterns. Chart patterns like the Head and Shoulders, Double Top, Inverse Head and Shoulders, and Double Bottom are considered the most reliable signals for identifying a change in market trend [1].

According to classic theories, these chart patterns reflect the exhaustion of buyers or sellers after a prolonged trend. However, identifying these chart patterns visually is

often subjective because they span multiple trading sessions and have inconsistent shapes. This led to the demand for automated systems based on AI to standardize recognition [7]. However, translating these concepts into robust computational models remains challenging because of the variability, noise, and structural complexity of real-world financial data. In addition to price structures, trading volume is widely recognized as a key factor in confirming market trends. Murphy [1] have said that "Volume should go with the trend. In an uptrend, volume should be heavier as the price moves higher, and should diminish on price dips." This principle highlights the important role of volume. This shows that price patterns should be interpreted together with volume behavior.

2.2 Success of YOLO in Financial Pattern Recognition

YOLO models have shown strong performance in object detection tasks. They have recently been applied to financial chart analysis. The study by Nguyen et al [5] provides clear evidence by using advanced models such as Scaled-YOLOv4, YOLOX, and YOLOR to detect complex chart patterns. The results show that YOLOX achieved a mean Average Precision (mAP@50) of up to 91.7% for Head and Shoulders patterns, Scaled-YOLOv4 reached 90.6% for the inverse pattern.

These figures confirm that YOLO is suitable for the Object Detection task in finance, helping to solve the latency problem of traditional methods and having good noise handling capability due to combination with segmentation techniques. Other applications, such as Birogul et al [8] , using YOLOv3 to achieve an 85% success rate in Buy/Sell decision-making, Santos et al [9] integrated Moving Averages into input images, enabling YOLOv8 to filter out false positives more effectively than relying just on pure price action. Their direct comparison revealed that YOLOv3 achieved a modest Recall of 0.69 (consistent with Temur's findings), the YOLOv8-MA approach significantly improved performance with a Recall of 0.87 and an F1-Score of 0.88. Overall, these studies demonstrate the effectiveness of YOLO-based models in detecting complex chart patterns under various market conditions, particularly due to their ability to process visual features in real time. Table 1 summarizes representative studies applying deep learning and object detection techniques to financial chart pattern recognition.

Table 1. Summary of related work on candlestick pattern detection using deep learning.

Study	Dataset	Period	Model	Key Results
[9]	Stocks: NASDAQ high-volatility stocks (AAPL, TSLA, NVDA, etc.). Input: Candlestick charts with Moving	2019 – May 2025	YOLOv8, YOLOv9, YOLOv11, YOLOv3.	*Precision: 0.88* *Recall: 0.87* *F1-Score: 0.88 (YOLOv8 outperformed YOLOv3*

	Averages (10 & 20 periods).			*significantly in recall).*
[10]	Stocks: AAPL. Crypto: Bitcoin (BTC). Forex: EUR/USD. Input: Daily candlestick images.	2017 – Oct 30, 2023	Pure CNN (VGG16 backbone). Tested YOLOv8 for pattern detection ("Include-Pattern" approach).	Accuracy: 70% F1-Score: ~0.699 *Key*
[5]	Stocks & Forex: 30 indices and 30 currency pairs from the Vietnam Stock Exchange. Input: Images of chart patterns (Head & Shoulders, Double Top, etc.).	Not specified. (Window size: 1 week).	Object Detection Models: YOLOX, Scaled-YOLOv4, YOLOR, EfficientDet, DETR.	mAP@50 (Mean Avg Precision): - YOLOX: 91.7% (Head & Shoulders) - Scaled-YOLOv4: 90.6% (Reverse H&S) - Overall: 88.19% (using segmentation).
[8]	Stocks: Borsa İstanbul (BIST) stocks. Input: Annual 2D candlestick charts.	2000 – 2018	YOLOv3	Success Rate: ~85% (84.6% decision success). Profit: 102.68% profit generated in a 376-day simulation.

These studies report high detection accuracy when using different YOLO models. Most studies focus on price-based chart images. They do not use trading volume. The report of Duong et al [10] provides an important insight into using YOLO for detection. When candlestick patterns detected by YOLOv8 were integrated into a market strength prediction model, the overall accuracy was around 70%. This result was not so different from just using CNN. This result shows that using YOLO alone was not necessary. Despite these promising findings, several challenges remain. Many studies rely on price-only representations, which may reduce model robustness under noisy market conditions. A few studies that compare multiple models within a consistent

experimental framework. This makes it difficult to identify the most effective model in detecting candlestick charts.

Because a comprehensive comparison of modern YOLO models using a price combine with trading volume representation is still limited, especially in Vietnam. So we decided to evaluate multiple modern models within a unified framework and to examine how volume information can improve the performance.

3 The proposed model

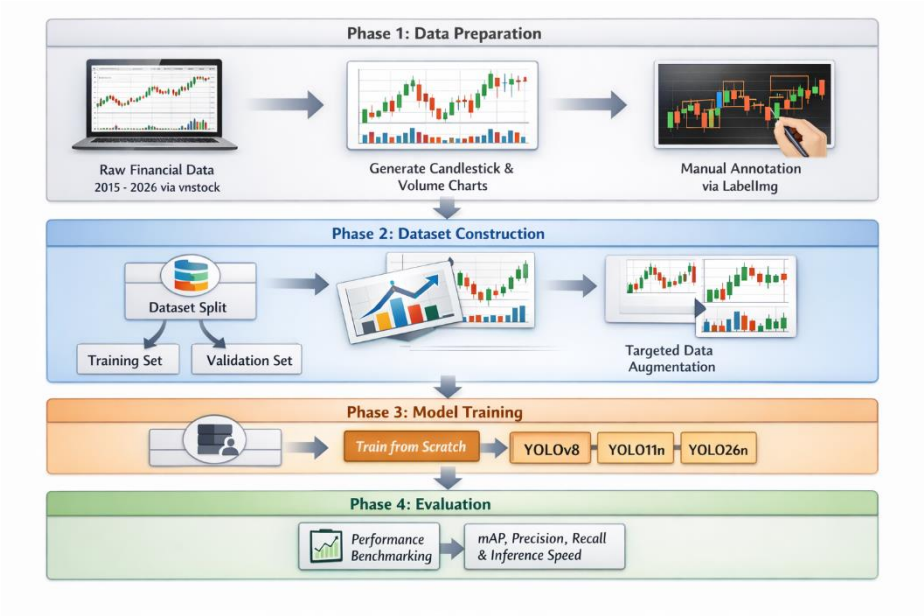

Fig. 1. Architecture of Proposed Model

Phase 1: Data Acquisition and Preprocessing

We used the vnstock library to collect historical stock market data in Vietnam. Data is collected from January 1, 2015, to February 6, 2026. The raw data includes important financial information like open, high, low, close prices (OHLC), trading volume, adjusted prices, and transaction values. To convert the numerical data into a suitable format for computer vision, we create high-resolution Candlestick and Volume charts. Using LabelImg [11], we manually label images with four chart pattern types: Head and Shoulders (HSA), Inverse Head and Shoulders (RHSA), Double Top (DT), and Double Bottom (DB).

Phase 2: Dataset Construction and Augmentation

Following image labeling, the data are separated into training and validation sets. We must apply data augmentation techniques to the training set in order to prevent overfitting and imbalance because the Double Top and Double Bottom patterns share few commonalities. To produce various market conditions, these strategies include brightness adjustment, noise addition, and scaling.

Phase 3: Multi-Model Training

We carry out a comparative study by training three different object detection models: YOLOv8, YOLO11n, and YOLO26n and set up the same hyperparameter for all.

Phase 4: Evaluation and Benchmarking

The final phase included a performance review. We compare three models with important metrics such as mAP@0.5, Precision, Recall, and inference speed (ms/image). This comparison aims to find the best balance model between detection accuracy and real-time processing efficiency for trading systems.

4 Experimental Results

4.1 Data Collection

This study uses historical stock price data from companies in the VN30 index listed on the Vietnam stock market. The data is collected using the vnstock library from January 1st, 2015, to February 6th, 2026. The collected historical data includes the following financial indicators: open price, close price, high price, low price, and trading volume. These indicators are used to create candlestick charts for the purpose of this research.

In total, we collected 71,356 observations from 30 stocks. After the candlestick charts are created, they are divided into sliding windows with a length of 60 trading days for each window. The 60-day period is selected to ensure that medium-term chart patterns can be fully observed within a single image. As a result, a total of 4,665 images is generated. These images are then manually labeled by the research team.

More than 600 chart patterns are manually labeled in this study. The labeled patterns include Head and Shoulders, Inverse Head and Shoulders, Double Top, and Double Bottom. These labels serve as input ground truth for the models in the next part of the study. Manual labeling is performed to ensure accurate pattern recognition and improve the learning quality of the models.

After manual labeling, the dataset was divided into two parts: 400 images for the training set and the remaining 200 for the control set. We also ensured the number of patterns in each set was proportionally balanced. To increase the size and diversify the input, thereby improving the performance of the models, enhancement techniques such as scaling, adding noise, and adjusting brightness were used in the initial training set. After enhancement techniques, the training dataset was increased to over 8000 images, which helps the models learn from more scenarios and reduces overfitting during the learning process.

Finally, these two improved datasets were used to train and evaluate the performance of YOLO models in a study for identifying stock market patterns.

4.2 Results & Comparative Analysis

Three object detection models - YOLOv8n, YOLO11n, and YOLO26n - were taken into consideration for this investigation [12]. 200 chart images with 245 occurrences in four pattern classes make up the validation dataset. Precision, Recall, and Inference Speed are used in conjunction with mAP@0.5 (Mean Average Precision at IoU=0.5), a popular metric for object detection, to provide a comprehensive assessment across all models.

The performance metrics for each model are compiled in Table 2. YOLOv8n has the highest overall accuracy (mAP@0.5 = 85.6%), which is interesting. Additionally, it is significantly higher than YOLO26n (77.0%) and marginally better than YOLO11n (85.3%).

Because those mAP@0.5 are above 90%, all three models exhibit good results for Head and Shoulders (HSA) and Inverse Head and Shoulders (RHSA) patterns. For the HSA class, YOLOv8n achieves a peak accuracy of 95.4%, but Double Top (DT) and Double Bottom (DB) were harder to find.

The numbers showed that YOLOv8n continued to achieve positive results for both DT (79.6%) and DB (77.2%), while YOLO26n showed a significant drop in performance, achieving only 55.0% on the DT class. In contrast to the more obvious structures like HSA/RHSA, this discrepancy illustrates the limitations of the training data, making it difficult for the models to identify dual-peak patterns.

Table 2. Comparison of AP@0.5 across models per pattern class.

Model	HSA (%)	RHSA (%)	DT (%)	DB (%)	mAP@0.5 (%)
YOLOv8n	**95.4**	90.4	**79.6**	77.2	**85.6**
YOLO11n	93.3	**94.3**	73.1	**80.6**	85.3
YOLO26n	92.2	87.9	55.0	73.0	77.0

We must examine the Precision and Recall in Table 3 in order to delve further into the output of the top model, YOLOv8n. The findings provide an intriguing narrative. When identifying Double Top patterns, the precision and recall are 88.3% and 57.1%, respectively. In other words, the model has a high false-negative rate and misses a lot of subtle DT patterns, but it also rarely generates false alarms.

Double Bottom tells a different story. Here, YOLOv8n prioritizes Recall (77.4%) over Precision (69.6%). The numbers suggest that the model is more aggressive in identifying bottoms, but it also generates more false positives by labeling random noise as valid patterns.

Table 3. Precision (P) and Recall (R) analysis per model.

Model	Metric	HSA (%)	RHSA (%)	DT (%)	DB (%)
YOLOv8n	**P**	83.7	75.5	**88.3**	69.6
	R	95.9	90.6	57.1	77.4
YOLO11n	**P**	86.3	76.9	83.7	73.0
	R	93.3	93.4	57.1	80.6
YOLO26n	**P**	67.3	62.5	77.4	58.7
	R	97.3	96.9	36.8	68.9

YOLO26n is the undisputed winner in terms of speed (Table 4), processing an image in a mere 3.7 ms, or roughly 270 FPS. However, speed isn't everything. There is a discernible decline in accuracy with that raw performance. In contrast, YOLOv8n finds its sweet spot at 4.8 ms per image, or about 208 FPS. That is more than sufficient for high-frequency trading or real-time market monitoring.

YOLOv8n is the most popular dataset. Overall, it performs well, but the DT and DB statistics point to a more serious issue: class imbalance and the great variation among pattern types.

Table 4. Average Inference Speed and Frame Rate.

Model	Speed (ms/img)	FPS
YOLOv8n	4.8	208
YOLO11n	6.0	166
YOLO26n	**3.7**	**270**

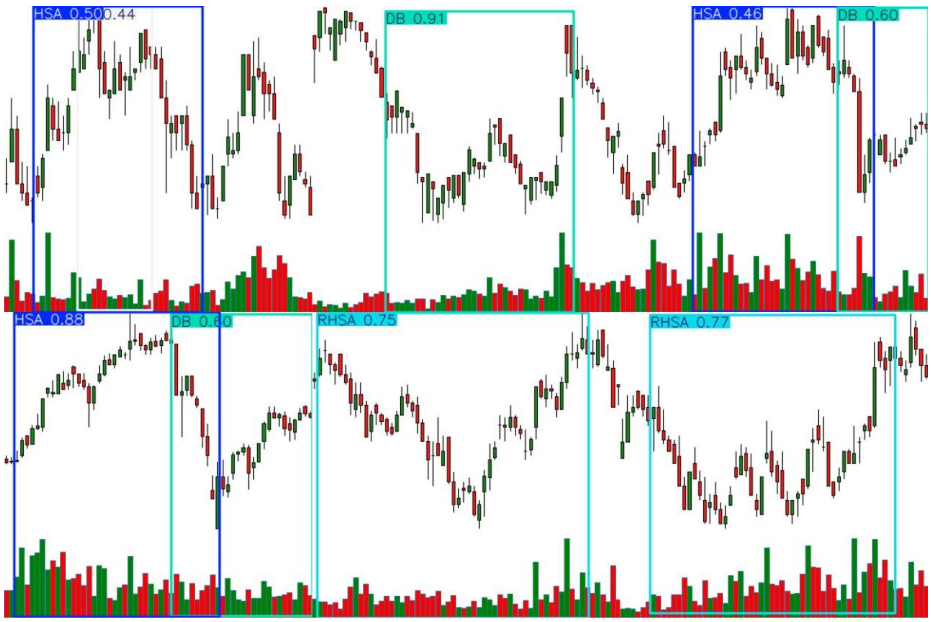

Fig. 2. Detection results of four candlestick patterns using YOLOv8n.

5 Conclusion

This study trained and evaluated three models, YOLOv8n, YOLO11n, and YOLO26n, on the same train and validation dataset to ensure fairness in model comparison. Two performance metrics were selected for comparison: accuracy and recall. The evaluation results were somewhat surprising to us. YOLOv8n was the best-performing model in the pattern recognition problem we posed, despite being the earliest of the three. YOLO11n closely followed in performance, lagging behind by only 0.3% in mAP, but its problem lay in its slowest processing speed among the three models. Conversely, YOLO26n excels in processing speed, living up to its initial launch claims, but at the cost of accuracy, as it has the worst recognition results. Perhaps this is because it is a newly developed model and needs more time to optimize its performance.

The advantage of this study lies in how I improved the representation of input data. Unlike previous studies that only considered price indicators, we also included trading volume data in the input image data. Based on John Murphy's technical analysis theory [1], patterns are only fully confirmed when accompanied by confirmation signals from trading volume. That is why we included the trading volume indicator in this study. Through training models to identify patterns along with trading volume, the research team showed that current advanced YOLO architecture models are capable of identifying much more complex financial patterns. The ability to identify patterns along with trading volume signals contributes to increased confidence among traders and ensures that detected patterns conform to the dynamics of the overall market. In addition, the inclusion of trading volume in the input of the models reduces the risk of false positives in models based solely on price indicators [13].

Besides the successes achieved, the study still faces unexpected challenges and needs improvement, most notably the imbalance of input patterns, especially in Double Top and Double Bottom patterns. These patterns are difficult to identify because they are often obscured by noise, which explains why the recognition performance of the models for these two patterns is significantly lower than that of the Head and Shoulders pattern and its inverse variant. We hope that future studies will have improvements to overcome limitations such as enriching data for the two unbalanced patterns, improving loss functions to better capture rare reversal signals [14]. In addition, this study is only focusing on price reversal patterns, not exploring continuation patterns which also play a very important role in the financial market, this will be a research gap for us or future works to continue researching and developing.

With the successes achieved, this research lays the foundation for integrating machine learning and deep learning models into the field of quantitative finance. With the ability to process huge datasets at high speed and with increasingly improved accuracy, these recognition models help traders minimize human psychological bias in considering financial patterns. Furthermore, future models promise not only to recognize patterns but also to understand the market mechanisms behind them [15, 16].

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